Highlights

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ABSTRACT

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1. Introduction and literature review

Analysing public transport supply from observed operation can be a very complicated task for governmental entities. The quantities that determine whether capacity is enough, such as peak loads and the occurrence of full buses, are extreme values, and estimating them reliably requires a large number of observations taken under the same conditions [1]. The binding constraint, however, is not the volume of data itself but the length of the period over which the service configuration remains unchanged. Each modification of the supply thus restarts the sample. For this problem, multiple entities rely on simulation techniques to analyse supply without real deployment [2].

This work focuses on the capacity of buses to absorb demand studying a digital simulation of a bus route in Madrid, Spain, using a multi-agent system.

The multi-agent paradigm is a well-known and widely implemented technique in transport simulation, and its use has increased in recent years [3–5]. Despite this, in public transport, the main area of study in which it applies is mostly real-time control systems [6], whereas this work focuses on demand level analysis.

Some of the works among the literature of MAS in transportation take advantage of the cognitive contribution certain paradigms such as BDI (Belief-Desire-Intention) offer, like modeling drivers and evacuees who must take decisions during a simulation [7, 8]. Such architectures are appropriate when the decision process of each individual agent is the object of study; when the focus is the aggregate behaviour of the system, simpler agents are enough [9]. The architectural decision of which types of agents to use for a certain MAS depends on the purpose the simulation aims at and on the set of data available.

In terms of granularity and nature of the simulation, different types appear in literature: macroscopic, mesoscopic and microscopic [10]. Microscopic simulations refer to those which describe the individual exact behaviour of entities, with a level of detail up to second by second. On the other hand, macroscopic simulations do not represent that individual behaviour, but rather use a global perspective to obtain aggregated level indicators and without the need to model invidividual entities. In the middle of those appear the mesoscopic. While it is used to represent aggregated level indicators, it still relies on using individual entities as so, but without the need of having them modelled at detail [2].

The mesoscopic approach seems to suit better for the purpose of this work, as each bus is represented as an entity, but its movement among the route comes from sampled data from fitted distributions and does not change by traffic dynamic inside the simulation. A microscopic level of detail is something this kind of problem does not require [2].

In this work passengers are used to analyse bus capacity, and the origin and destination of individual entities are not required. No external surveys or other data from which to construct OD matrices are available. The usual approach is to infer the matrices via IPF with null seeding [11, 12], while in this work the boarding and alighting of passengers is done via fitted distributions from real data operations. This decision comes from the goal of the work. OD matrices assigns origin and destination to passengers, but here the problem requires only to know how many boards and alightings occurs in a specific stop given a bus arrival. Several bus simulators already operate from stop-level counts without an

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OD matrix. BusMezzo [2] is a mesoscopic simulator in which passengers are not represented individually. Arrivals at stops follow a Poisson process, and the number of alighting passengers is drawn from those on board. BusSim [13] is an agent-based model with bus and stop agents in which passengers represents aggregate quantities, built on synthetic data for short-term prediction. Talusan et al. [14] constructs an event-based simulator with APC (Automatic Passenger Counting) data to evaluate dispatching decisions. These works establish that occupancy can be studied without passenger destinations. However, first and third works are not multi-agent systems, the second is not calibrated on real operation, and none of them models the same line with and without individual passengers.

For the agents used in this paper, as the work studies no individual decision and the available data contain no information about one, no cognitive reasoning is represented for either the bus or the passenger. The agents are therefore reactive in the classification of Wooldridge and Jennings [15]: they act in response to events and to the messages they receive. They are not purely reactive, however, as they keep an internal state [16]. The form they have of interaction is through FIPA interaction protocols []. The communication between different entities is the core of the behaviour of the system.

From the engineering side, this work provides a multi-agent model of line E01 of EMT Madrid, calibrated with the boardings and alightings recorded by the operator and used to analyse occupancy and demand absorption. From the artificial intelligence side, it addresses how fine the agents of such a system need to be. Two solutions are built that share the same bus model and the same interaction protocols between reactive agents, and differ only in whether each passenger is an agent or passengers are counts held by stops and buses. Comparing them isolates what the passenger agent contributes, in results and in computational cost, when the available data record only counts.

The remainder of the paper is organised as follows. Section 2 presents the two proposed solutions, defining the agents and the interaction protocols between them. Section 3 describes the operational data and how the boarding and alighting distributions are calibrated from them. Section 4 details the implementation in GAMA, the simulation parameters for line E01 and the results obtained. Finally, Section 5 draws the conclusions.

2. Proposed solutions: two different approaches

As for the purpose of this work, no cognitive reasoning is represented neither for the bus nor the passenger. Thus, the emergent intelligence comes from the interaction between agents and environment, as per Brooks [17], that emergent behaviour guarantees an agency right itself.

2.1. Agents and protocols

As stated before, the agents that are modelled in this work are reactive agents under the wooldridge and Jennings taxonomy, but they kept themselves an internal state, which gives them not the full reactivity core, but more

In the two proposed implementations, three agents are defined:

- bus
- stop
- Passenger

The protocols are defined in figures 1, 2.

3. Data calibration

The system uses parametric data obtained through analysis of empirical 8 month operational window timespan.

The data covers all stop related information for every time in a day a bus reaches a stop.

In this work, as per cardinality constraints due to not reaching enough statistical data for covering all scenarios, only the labour days are included in the parametric analysis. Moreover, the freeday data is also excluded from the analysis, as it changes the behaviour of the buses occupation. The time series anal

3.1. Attributes for calibration

The system is modelled for receiving four attributes.

For every stop, disaggregated into its timeslots, it expects distribution fitted parameters for both boarding and alighting processes.

The boarding value represents the number of people arriving to a stop each hour, and it will use it as a passengers generator ratio.

On the other hand, the alighting values are modelled as independent samples per bus arrival. Each time a bus reaches a stop, the system will pick a random sampling out of the distribution corresponding to the combination (stop-timeslot).

Things to mention: Why only labour, why no freeday, why no possibility of decouple stationality and analysis via temporal series

Also, mention the chosen distribution families, the why use of negbins and geoms, cause the variance > mean (poisson is variance == mean) Also, mention the NO OD data, and as i chose not to infer with IPF NULL seeding, its constructed by independent consideration of boarding and alighting.

3.2. Parametric implementation

4. Implementations and results

Both proposed architectures run in the dedicated agent platform GAMA [18]. The election is based on both the inclusion of FIPA inboxing, allowing implementation-defined conversations, and a headless batch mode that improves performance across runs.

4.1. Simulation parameters

For the simulation characterization, the layout of route Express 01 is used. The route operates between the central square *Cibeles*, located in the city center of the Spanish capital, and *La Peseta*, located in the southern neibourhood of Carabanchel. The route has 15 and 12 stops for each direction respectively, and its core behavour reflects the day by day of madrid's workers: Going to the center earlier in the morning to work and coming back home after lunch. The route's prime time in Cibeles's direction comes between 7 and 9 in the morning, boarding passengers in the first stops of the trip, and alighting them at the final ones. The same behaviour is inversed in La Peseta's direction, presenting its prime time after 3 pm, capturing the tendency of workers to return to its homeplace.

The specific parameterization used is shown in table 1.

Table 1
Simulation parameters.

Parameter	Value
Line	E01 (Cibeles–La Peseta)
Physical buses	12
Trips per day	163
Initial time slot	5
Extra dwell time per passenger (t_p)	5 s
Simulation time step (Δt)	2 s
Minimum protocol dwell ($E_{\min}$)	$4\,\Delta t$
Monte Carlo replicas	1000

Before the results are presented, some specifications about the simulations itself are introduced. The field `Extra dwell time per passenger` is used with both a premise and a comparison.

First, it assumes that the boarding and alighting processes happen in parallel, as in regular bus operations those occur in different physical doors.

Secondly, the dwell extra time (E) will be the result of the following equation:

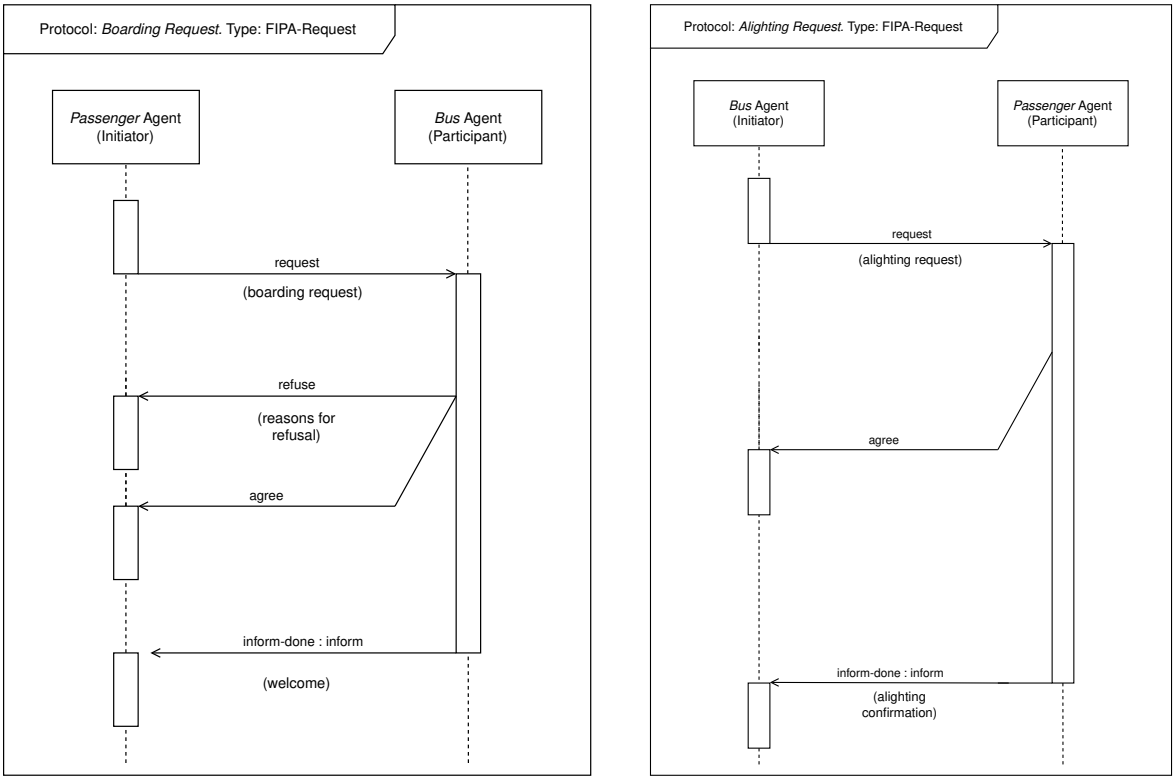
$$E = \max\big(\max(b,\ a)\, t_p,\ E_{\min}\big) \tag{1}$$

where b and a are the numbers of boarding and alighting passengers at the stop, t_p is the extra dwell time per passenger, and $E_{\min}$ is the minimum dwell imposed by the interaction protocol.

In relation to the necesary input files required to run the simulation,

4.2. Results

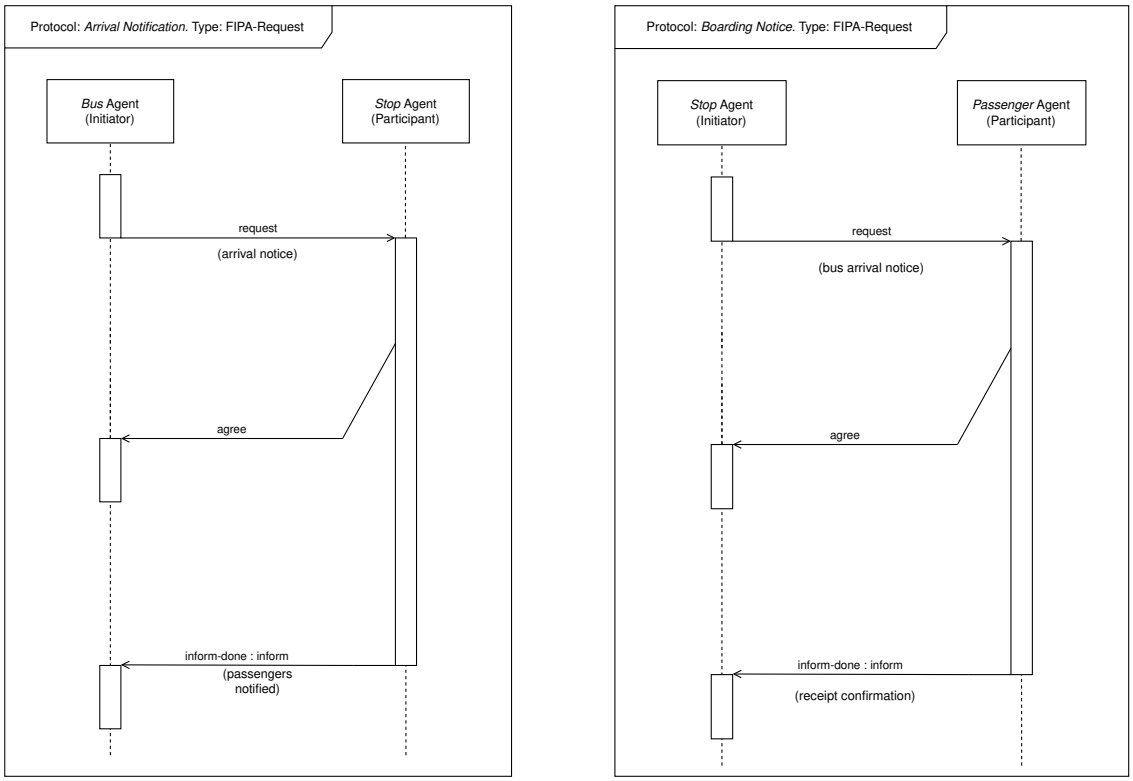
5. Conclusions



(a) Boarding request

(b) Alighting request

Figure 1: FIPA-Request interaction protocols for passenger requests.



(a) Boarding request (b) Alighting request

Figure 2: FIPA-Request interaction protocols for a bus arriving to a stop.

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CRediT authorship contribution statement

H. Álvarez: Aqui va la contribucion mia. **J. Carbó:** Aqui va la contribucion de Javier/José Manuel. **J.M. Molina:** Aqui va la otra contribucion de Javier/José Manuel.

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